

2-5 Addition Patterns

Learning goal: I can use even and odd patterns to tell whether a sum is even or odd, explain why, and use the patterns to check a sum.

Textbook: pages 51–54. The book shows the three even/odd sum patterns with 3-digit examples.

Words to Know

even number	A number that can be split into pairs with none left over. Its ones digit is 0, 2, 4, 6, or 8.
odd number	A number that has one left over when you make pairs. Its ones digit is 1, 3, 5, 7, or 9.
pattern	Something that happens again and again in the same way.
ones digit	The digit in the ones place. In 427, the ones digit is 7.
sum / addend	The answer when you add / a number that you add.

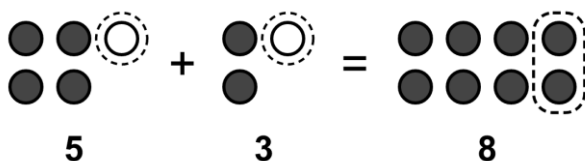
How to Teach It

1. Use small objects. Make pairs. If one is left over, the number is odd. If none are left over, it is even.
2. Join two odd groups: the two leftovers make a new pair, so **odd + odd = even**. Even + even has no leftovers, so it is **even**. Even + odd still has one leftover, so it is **odd**.
3. For big numbers, look only at the **ones digits**. Tens and hundreds are made of groups of ten, and ten is even, so they never leave a leftover.
4. Before adding, predict: will the sum be even or odd? After adding, check that the sum matches.
5. Remember: if the sum does not match the pattern, it is wrong. If it does match, you still need to add to be sure.

Tip: Ask, “What is the ones digit? Will there be a leftover?”

Worked Examples

Example 1: Why odd + odd = even



5 and 3 each have one leftover (dashed). Together the leftovers make a pair, so 8 is **even**.

Example 2: The three patterns

even + even = even $214 + 352 = 566$

odd + odd = even $135 + 221 = 356$

even + odd = odd $412 + 235 = 647$

The order does not matter: odd + even = odd too.

Example 3: Only the ones digits matter

$627 + 154$: look at $7 + 4 = 11$, which is odd. So the sum is **odd**.

Add to find it: $627 + 154 = 781$. 781 is odd. ✓

Example 4: Check a sum

Is $316 + 151 = 457$ correct?

even + odd = odd, and 457 is odd, so the pattern matches. But we must still add: $316 + 151 = 467$. The pattern can catch some mistakes, not all.

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|---|--|
| ☐ Tell if a number is even or odd from its ones digit. | ☐ Write a 3-digit equation that fits a pattern. |
| ☐ Say the three sum patterns. | ☐ Predict if a sum is even or odd, then add to check. |
| ☐ Explain the patterns with pairs and leftovers. | ☐ Know that a matching pattern does not prove a sum is right. |

2-5 Addition Patterns

Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Write **even** or **odd**. Then write one equation with 3-digit numbers that fits.

odd + odd = _____

Equation: _____

- B.** Is the sum of **358 + 227** even or odd? Decide before you add. Then find the sum.

Even or odd: _____

Sum: _____

- C.** Omar writes $243 + 316 = 549$. He says it is correct because $\text{odd} + \text{even} = \text{odd}$. Do you agree? Explain.

_____**On My Own** *Try these by yourself. Show your work.*

- 1.** Write **even** or **odd**. Then write two equations with 3-digit numbers that fit.

_____ = **odd + even**_____

- 2.** Write **even** or **odd**. Then write two equations with 3-digit numbers that fit.

even + even = __________

- 3.** Write **even** or **odd**. Then write two equations with 3-digit numbers that fit.

even + _____ = **odd**_____

- 4.** Write **even** or **odd**. Then write two equations with 3-digit numbers that fit.

odd + _____ = **odd**_____

5. Write **even** or **odd**. Then write two equations with 3-digit numbers that fit.

_____ + odd = even

6. Write **even** or **odd**. Then write two equations with 3-digit numbers that fit.

_____ = odd + odd

7. Find the sum. Use a pattern to check your answer.

$$352 + 417 = \underline{\hspace{2cm}}$$

8. Find the sum. Use a pattern to check your answer.

$$263 + 525 = \underline{\hspace{2cm}}$$

9. Why is the sum of a number with a **5** in the ones place and a number with a **2** in the ones place always odd?

10. A pet shop has two fish tanks. One tank has 25 fish. How many fish could be in the other tank so that the total number of fish is **even**? Explain.

11. Ava reads 121 pages, then 122 pages, then 124 pages. Is the total number of pages even or odd? Explain how you know.

2-5 Addition Patterns

Together

A	even. Sample: $135 + 241 = 376$ <i>Accept any two odd 3-digit numbers with a correct sum.</i>
B	Odd ($8 + 7 = 15$, and 15 is odd). Sum: 585 <i>even + odd = odd.</i>
C	No. The pattern is right (the sum should be odd), but $243 + 316 = 559$, not 549. <i>A sum can match the pattern and still be wrong. You have to add to be sure.</i>

On My Own

1	odd. Samples: $347 = 133 + 214$; $527 = 405 + 122$ <i>For items 1–6, accept any 3-digit equations that fit the pattern and are added correctly.</i>
2	even. Samples: $312 + 124 = 436$; $250 + 406 = 656$
3	odd. Samples: $132 + 215 = 347$; $400 + 301 = 701$
4	even. Samples: $213 + 104 = 317$; $121 + 340 = 461$
5	odd. Samples: $123 + 211 = 334$; $305 + 105 = 410$
6	even. Samples: $111 + 223 = 334$; $455 + 135 = 590$
7	769. even + odd = odd, and 769 is odd.
8	788. odd + odd = even, and 788 is even.
9	$5 + 2 = 7$, which is odd. The tens and hundreds are groups of ten, which are even, so only the ones digits decide. <i>Also accept a pairs explanation: 5 has one leftover and 2 has none, so one is left over.</i>
10	Any odd number , such as 13 ($25 + 13 = 38$) or 25 ($25 + 25 = 50$). <i>25 is odd, and odd + odd = even.</i>
11	Odd. $121 + 122$ is odd + even = odd. Then odd + 124 (even) = odd. Total: 367. <i>Also accept checking the ones digits: $1 + 2 + 4 = 7$, which is odd.</i>